GOVERNANCE · DOMAIN 8 - DATA GOVERNANCE

Unity Catalog permission inheritance

Section 8. Three things must be true, and ownership is not one of them.

SECTION 8 of 10 WEIGHT 7% TOPIC Permissions

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To read a table, three separate things must be true. Two of them are on objects that are not the table.

Containers and everything else

- A metastore at the top, then a **three-level namespace**: catalog, schema, table.
- **Container objects** are **catalogs**, which hold schemas, and **schemas**, which hold tables, views, volumes and functions.
- Tables, views, volumes and functions are **non-container**. Nothing lives inside them.
- Only containers do inheritance, require usage privileges on themselves, and support creation privileges.

> ON THE EXAM
>
> The container distinction is not decoration. It determines which objects can pass privileges down and which cannot.

What each operation actually requires

OPERATION	PRIVILEGES NEEDED
Read a table or view	USE CATALOG, USE SCHEMA, and SELECT
Write to a table	USE CATALOG, USE SCHEMA, and MODIFY
Create a schema	USE CATALOG, and CREATE SCHEMA on the catalog
Create a table	USE CATALOG, USE SCHEMA, and CREATE TABLE on the schema
Execute a function	USE CATALOG, USE SCHEMA, and EXECUTE
Read files from a volume	USE CATALOG, USE SCHEMA, and READ VOLUME

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Notice the shape. Two usage privileges on the parents, then one specific privilege on the object. Learn the shape rather than the rows.

Usage privileges are an administrative boundary

- A table owner can grant SELECT on their table to anybody. That grant alone does nothing.
- The recipient still cannot read it without **USE CATALOG and USE SCHEMA** on the parents.
- Only **catalog and schema owners, or MANAGE holders**, can grant those usage privileges.
- So the design **prevents table owners from granting access outside approved boundaries**.

> ON THE EXAM
>
> This is stated as a purpose, not a side effect. Usage privileges exist so that higher-level administrators keep control of the perimeter.

Current and future, downward only

- A privilege on a **catalog** applies to every schema in it, and every table, view, volume and function in those schemas.
- A privilege on a **schema** applies to everything in that schema.
- It covers **current and future** objects. A table created tomorrow is already covered.
- That is powerful and blunt. Granting SELECT on a catalog grants it on tables nobody has designed yet.

> ON THE EXAM
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> The future clause is the point of inheritance and its main risk. Grant broadly and you have granted on things that do not exist.

Metastore grants do not inherit

- **Privileges granted on a metastore do not inherit to child objects.** The chain starts at the catalog.
- Metastore grants cover metastore-scoped operations: create catalog, create external location, create share.
- There is **one documented exception**. **READ METADATA on a metastore does inherit** to every object in it.
- And that exception **needs no usage privileges** to exercise, unlike everything else.

> ON THE EXAM
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> So the highest level privileges inherit from is the catalog, with a single named exception for metadata visibility.

Implied, not granted

- The principal that creates an object becomes its **initial owner**. An owner may be a user, service principal or group.
- Owners have all capabilities, and **ALL PRIVILEGES is not explicitly granted to them**.
- So **SHOW GRANTS will not return ALL PRIVILEGES for the owner**. The capability is implied.
- Owners also need **no usage privileges** at all.

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> An empty grants list on an object you own is expected, not a misconfiguration. That distinction catches people auditing permissions.

Ownership does not inherit downward

- **Owning a catalog does not make you the owner of its schemas.** Ownership does not flow down.
- But an owner **automatically gets the ability to manage all child objects**, current and future.
- That too is **implied rather than explicitly granted**, so it will not show in a grants listing either.
- Manage is not the same as read. A catalog owner can administer a table without being able to query it.

> ON THE EXAM
>
> Two things are true at once: you do not own the children, and you can manage them. Questions are built on people assuming only one of those.

Ownership against the MANAGE privilege

OWNER

Capabilities implied

- All capabilities on the object, implied
- Manage child objects, implied
- No usage privileges required
- Exactly one principal

MANAGE

Administration without access

- Privileges must be granted separately
- But holders can grant them to themselves
- Usage needed only above where it is held
- Can be granted to many principals

THE TIEBREAKER

The practical difference: MANAGE lets several people administer an object, and ownership is a single principal that must be transferred.

MANAGE has reduced usage requirements

- You need usage only on levels **strictly above** where MANAGE is held, not at or below it.
- **MANAGE on a catalog requires no usage privileges at all.**
- MANAGE on a schema requires USE CATALOG on the parent, but not USE SCHEMA on the schema.
- MANAGE on a table requires both USE CATALOG and USE SCHEMA.

> ON THE EXAM
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> Because MANAGE on a container is inherited by its children, holding it high up means you need no usage privileges below.

Data access still needs usage privileges

- The reduction applies **only to metadata capabilities** that MANAGE provides.
- **SELECT and MODIFY still require USE CATALOG and USE SCHEMA**, even for a MANAGE holder.
- So MANAGE on a catalog lets you administer everything and read nothing, until you grant yourself more.
- That separation is deliberate. Administration and access are different jobs.

> ON THE EXAM
>
> A MANAGE holder who grants themselves SELECT also needs the usage privileges. The reduction never applies to reading data.

What ALL PRIVILEGES does not include

- It **implies** all applicable privileges for the object type, evaluated at permission-check time rather than at grant time.
- It does **not** include **MANAGE**. Excluded explicitly to avoid accidental privilege escalation.
- It also excludes **READ METADATA**, **EXTERNAL USE SCHEMA** and **EXTERNAL USE LOCATION**.
- Evaluating at check time means a privilege added to the platform later is covered automatically.

> ON THE EXAM
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> All does not mean all. Four named privileges sit outside it, three of them because they cross a security boundary.

The BROWSE privilege

- **BROWSE lets users discover objects and view metadata without any access to the underlying data.**
- They see the object exists, and its name, description and tags, and can **request access**.
- It works **without USE CATALOG or USE SCHEMA**, which is unusual and is the point.
- Granted at **catalog level**. Databricks recommends granting it on catalogs to **all account users**.

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> This is the bridge to the next module. Discoverability is a governance objective, and BROWSE is how you get it without granting data access.

Composite and child privileges

- **MANAGE** is composite, and **READ METADATA** is its child, giving read-only metadata visibility without the management abilities.
- **MODIFY** is composite, and **INSERT, UPDATE and DELETE** are its children.
- They are granted and revoked **independently**. Granting the composite does not grant the children.
- And **revoking the composite does not revoke a child that was granted explicitly**.

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> That second half is the trap. Revoke MODIFY from somebody who also holds DELETE explicitly, and they can still delete.

Two things owners deliberately do not get

- **Schema owners do not get EXTERNAL USE SCHEMA by default.**
- **External location owners do not get EXTERNAL USE LOCATION by default.**
- Both defaults exist to avoid **accidental data exfiltration**.
- So ownership is not quite unlimited. Two capabilities are withheld until somebody grants them deliberately.

> ON THE EXAM
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> An exception to the rule that owners have every capability, and the reason given is exfiltration rather than tidiness.

What to carry into the exam

- **Three things must be true:** USE CATALOG, USE SCHEMA, and the specific privilege.
- Inheritance runs **catalog downward, current and future**. **Metastore grants do not inherit**, except READ METADATA.
- **Ownership does not inherit**, but owners can manage all children. Both are implied, not granted.
- **ALL PRIVILEGES excludes MANAGE**, READ METADATA, and the two external use privileges.
- **BROWSE** gives discovery without data access, and without usage privileges.

> ON THE EXAM
>
> When a permissions question looks like it needs one grant, count again. It usually needs three.

> A table owner grants SELECT on their table to the analytics group so they can build a report. The analysts report that the table does not appear and they cannot query it.
>
> The grant is confirmed present with SHOW GRANTS on the table.

What is missing?

A. USE CATALOG on the catalog and USE SCHEMA on the schema. All three are required.

B. The analysts need BROWSE on the table before SELECT takes effect.

C. The grant must be made at schema level to be inherited by the table.

D. The table owner must transfer ownership to the analytics group.

◆ PAUSE AND DECIDE

What is missing?

A USE CATALOG on the catalog and USE SCHEMA on the schema. All three are required.

Stated directly: SELECT alone is not sufficient without usage privileges on both parents.

B The analysts need BROWSE on the table before SELECT takes effect.

BROWSE aids discovery and is granted at catalog level. It is not a prerequisite for SELECT.

C The grant must be made at schema level to be inherited by the table.

A schema-level grant would work, and only because it comes with usage on the parents in practice. The table grant is valid.

D The table owner must transfer ownership to the analytics group.

Enormous overreach for a read requirement, and there can only be one owner.

ON THE EXAM

The table owner cannot fix this themselves. Only catalog and schema owners, or MANAGE holders, can grant the usage privileges.

> A governance lead owns the finance catalog. They are asked to audit permissions and find that SHOW GRANTS on a table within it returns nothing for them, though they can administer the table.
>
> They also cannot query the table's contents.

Which explanation fits all three observations?

A Permissions are broken and need to be re-granted from the catalog down.

B Catalog owners get implied management of child objects, not explicit grants, and management is not data access.

C They must be a metastore admin to see grants on child tables.

D SHOW GRANTS only lists grants made in the last 90 days.

◆ PAUSE AND DECIDE

Which explanation fits all three observations?

A Permissions are broken and need to be re-granted from the catalog down.

Nothing is broken. Every observation is documented behaviour.

B Catalog owners get implied management of child objects, not explicit grants, and management is not data access.

Explains all three: implied capabilities do not appear in SHOW GRANTS, and managing is not reading.

C They must be a metastore admin to see grants on child tables.

Object owners can view grants. The point is that their own implied capability is not listed.

D SHOW GRANTS only lists grants made in the last 90 days.

Invented.

ON THE EXAM

Three observations, one mechanism. Implied capabilities are real but invisible, and management is separate from access.

> A security review requires that a contractor be removed from write access to a table. The contractor holds MODIFY on the table, and also holds DELETE, granted separately six months earlier.
>
> An administrator revokes MODIFY and closes the ticket.

What is the problem?

A Nothing. Revoking MODIFY revokes its child privileges.

B Composite and child privileges are revoked independently. The explicit DELETE grant remains, so they can still delete rows.

C Revoking MODIFY also removes SELECT, so the contractor now has no access at all.

D DELETE cannot be granted separately from MODIFY.

◆ PAUSE AND DECIDE

What is the problem?

A Nothing. Revoking MODIFY revokes its child privileges.

The docs state the opposite: revoking a composite does not revoke an explicitly granted child.

B Composite and child privileges are revoked independently. The explicit DELETE grant remains, so they can still delete rows.

Granting and revoking are independent in both directions. The child grant survives.

C Revoking MODIFY also removes SELECT, so the contractor now has no access at all.

SELECT is a separate privilege, not a child of MODIFY.

D DELETE cannot be granted separately from MODIFY.

It can. That is what a child privilege is for.

ON THE EXAM

Revoking the broad privilege and assuming the narrow ones went with it is exactly the wrong direction to be wrong in.

> A data platform team wants analysts across the company to be able to find datasets and request access, without being able to read anything they have not been granted.
>
> A proposal suggests granting SELECT on all catalogs to all account users, then relying on row filters to restrict what they see.

What is the better approach?

A The proposal is sound; row filters will restrict the data appropriately.

B Grant BROWSE on the catalogs to all account users, which allows discovery and access requests without data access.

C Grant USE CATALOG to all account users, which allows discovery without SELECT.

D Grant READ METADATA on each table individually.

◆ PAUSE AND DECIDE

What is the better approach?

A The proposal is sound; row filters will restrict the data appropriately.

Grants SELECT on all current and future tables, then relies on filters existing everywhere. One missing filter exposes a table.

B Grant BROWSE on the catalogs to all account users, which allows discovery and access requests without data access.

Exactly what BROWSE is for, and Databricks recommends this pattern by name.

C Grant USE CATALOG to all account users, which allows discovery without SELECT.

Closer, and BROWSE is the purpose-built privilege and needs no usage privileges.

D Grant READ METADATA on each table individually.

Would work and is per-object toil. BROWSE is granted once at catalog level.

ON THE EXAM

A is the dangerous one: it grants SELECT on future tables too, so a table created next year is exposed until somebody remembers a filter.

A governance team needs to administer permissions across an entire catalog: granting, revoking, and transferring ownership. Policy says they must not be able to read the data itself.

Three people need this capability.

An administrator proposes transferring catalog ownership to the governance lead.

What is the better arrangement?

A Transfer ownership to the lead, then have them delegate to the other two.

B Grant MANAGE on the catalog to all three, which requires no usage privileges and does not confer data access.

C Grant ALL PRIVILEGES on the catalog to the governance group.

D Make the governance group the owner, since a group can own an object.

◆ PAUSE AND DECIDE

What is the better arrangement?

A Transfer ownership to the lead, then have them delegate to the other two.

Ownership implies all capabilities including data access, which policy forbids.

B Grant MANAGE on the catalog to all three, which requires no usage privileges and does not confer data access.

MANAGE can be held by many, needs no usage privileges at catalog level, and does not imply SELECT.

C Grant ALL PRIVILEGES on the catalog to the governance group.

Confers data access, and notably does not include MANAGE.

D Make the governance group the owner, since a group can own an object.

True that a group can own, and ownership still implies all capabilities including reading.

ON THE EXAM

Note that C fails in both directions at once: it grants the data access policy forbids, and withholds the MANAGE they actually need.

> An auditor must be able to see the name, description and lineage of every object in the metastore, across all catalogs, without any ability to read data.
>
> A proposal is to grant them BROWSE on each catalog as it is created.

Is there a simpler option?

A. No. BROWSE per catalog is the only mechanism for metadata visibility.

B. Grant READ METADATA on the metastore, which is the one privilege that inherits from the metastore and needs no usage privileges.

C. Grant ALL PRIVILEGES on the metastore, which inherits to every object.

D. Make the auditor a metastore admin, which is the standard approach.

◆ PAUSE AND DECIDE

Is there a simpler option?

A. No. BROWSE per catalog is the only mechanism for metadata visibility.

BROWSE works and is per catalog. READ METADATA on the metastore covers everything at once.

B. Grant READ METADATA on the metastore, which is the one privilege that inherits from the metastore and needs no usage privileges.

The documented exception to metastore grants not inheriting, and it needs no usage privileges to exercise.

C. Grant ALL PRIVILEGES on the metastore, which inherits to every object.

Metastore grants do not inherit, and ALL PRIVILEGES excludes READ METADATA anyway.

D. Make the auditor a metastore admin, which is the standard approach.

An enormous grant to solve a read-only metadata requirement.

ON THE EXAM

The requirement says every object across all catalogs. That phrasing points at the one privilege that inherits from the metastore.